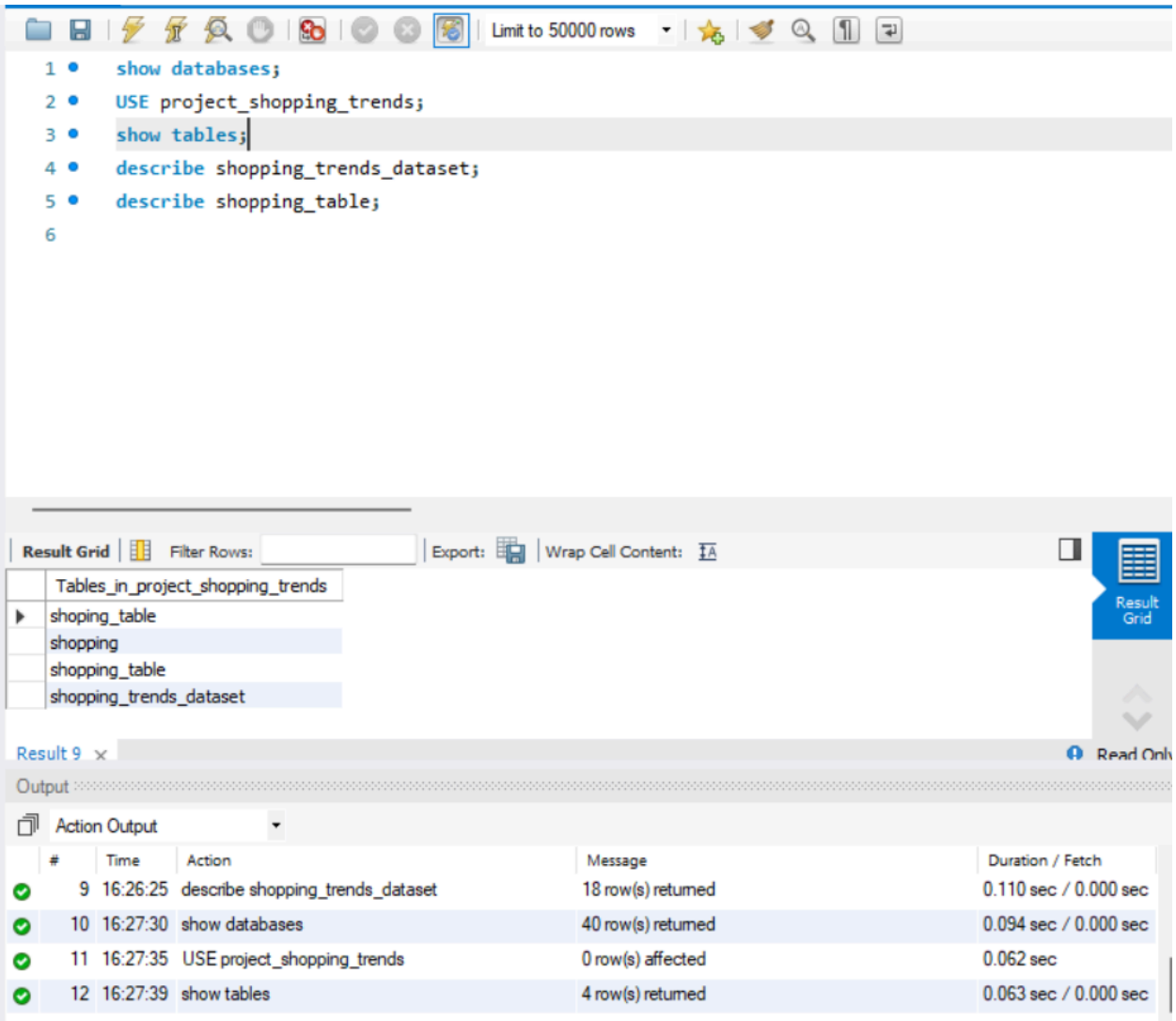
Client Project: project_shopping_trends

Based on purchase data from shoppers in the U.S. market, performed the following tasks:

1) Identify the product categories where discounts should be applied. Provide your reasoning.

2) Identify card spending based on age and explore the impact of seasons and locations.



1. Category wise discount analysis:

```sql
describe shopping_table;

# 1) Identify the product categories where discounts should be applied. Provide your reasoning.
# Category-wise Discount Analysis
SELECT Category,
        COUNT(*) AS total_orders,
        AVG(Purchase_Amount_USD) AS avg_amount,
        SUM(CASE WHEN Discount_Applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*) AS discount_rate
FROM shopping_table
GROUP BY Category
ORDER BY total_orders DESC;
```

Result Grid

Category	total_orders	avg_amount	discount_rate
Clothing	1737	60.0253	0.4208
Accessories	1240	59.8387	0.4379
Footwear	599	60.2554	0.4324
Outerwear	324	57.1728	0.4444

Result 15

Output

Action Output

#	Time	Action	Message
15	16:55:24	show tables	4 row(s) returned
16	16:55:31	describe shopping_table	19 row(s) returned
17	16:59:40	SELECT Category, COUNT(*) AS total_orders, AVG(Purchase_Amount_USD) AS avg_amount, SUM...	4 row(s) returned
18	17:00:04	SELECT Category, COUNT(*) AS total_orders, AVG(Purchase_Amount_USD) AS avg_amount, SUM...	4 row(s) returned

2. Card Spending by age:

```
ORDER BY total_orders DESC;

# 2)    Identify card spending based on age and explore the impact of seasons and locations.
# Card Spending by Age
SELECT Age,
       SUM(Purchase_Amount_USD) AS card_spend
FROM shopping_table
WHERE Payment_Method LIKE '%Card%'
GROUP BY Age
ORDER BY Age;
```

Result Grid

Age	card_spend
18	1258
19	1703
20	1166
21	1852
22	1309
23	1166
24	1220
25	2033
26	1234
27	1437
28	1852
29	1355

Result 18

Output

Action Output

#	Time	Action	Message
20	17:04:56	SELECT Age, SUM(Purchase_Amount_USD) AS card_spend FROM shopping_table WHERE Payment_...	53 row(s) returned
21	17:05:39	SELECT Age, SUM(Purchase_Amount_USD) AS card_spend FROM shopping_table WHERE Payment_...	53 row(s) returned

3. Season and Location impact:

```
GROUP BY Age
ORDER BY Age;

# Season & Location Impact
SELECT Season, Location,
       SUM(Purchase_Amount_USD) AS total_spend
FROM shopping_table
GROUP BY Season, Location
ORDER BY total_spend DESC;
```

Result Grid

Season	Location	total_spend
Spring	Nevada	1916
Winter	Vermont	1885
Spring	Alaska	1848
Summer	Pennsylvania	1753
Winter	Virginia	1752
Fall	Texas	1732
Fall	California	1659
Summer	North Dakota	1656
Winter	Montana	1631
Fall	Idaho	1622
Spring	Illinois	1620
Summer	Alabama	1610

Result 20

Output

Action Output

#	Time	Action	Message
22	17:15:34	describe shopping_table	19 row(s) returned
23	17:16:14	SELECT Season, Location, SUM(Purchase_Amount_USD) AS total_spend FROM shopping_table GROU...	200 row(s) returned

4. EDA + insights :

1. Count rows & columns
2. Column list
3. Missing values count per column

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Limit to 50000 rows

```sql
# EDA + insights
-- 1. Count rows & columns
SELECT COUNT(*) AS total_rows FROM shopping_table;

-- 2. Column list (MySQL)
SHOW COLUMNS FROM shopping_table;

-- 3. Missing values count per column (approx)
SELECT
  SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS age_null,
  SUM(CASE WHEN Gender IS NULL THEN 1 ELSE 0 END) AS gender_null,
  SUM(CASE WHEN Purchase_Amount_USD IS NULL THEN 1 ELSE 0 END) AS amount_null,
  SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS category_null,
  SUM(CASE WHEN Discount_Applied IS NULL THEN 1 ELSE 0 END) AS discount_null
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows
3900

Result 25

Output

Action Output

#	Time	Action	Message
28	20:24:28	SELECT Customer_ID, Age, Gender, Item_Purchased, Category, Purchase_Amount_USD, Location, Size,...	3900 row(s) affected
29	20:29:55	SELECT COUNT(*) AS total_rows FROM shopping_table LIMIT 0, 50000	1 row(s) returned

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Limit to 50000 rows

```sql
-- 3. Missing values count per column (approx)
SELECT
  SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS age_null,
  SUM(CASE WHEN Gender IS NULL THEN 1 ELSE 0 END) AS gender_null,
  SUM(CASE WHEN Purchase_Amount_USD IS NULL THEN 1 ELSE 0 END) AS amount_null,
  SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS category_null,
  SUM(CASE WHEN Discount_Applied IS NULL THEN 1 ELSE 0 END) AS discount_null
FROM shopping_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_null	gender_null	amount_null	category_null	discount_null
0	0	0	0	0

Result 27

Output

Action Output

#	Time	Action	Message
30	20:31:35	SHOW COLUMNS FROM shopping_table	19 row(s) returned
31	20:33:07	SELECT SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS age_null, SUM(CASE WHEN Gender...	1 row(s) returned

4. Category-level summary and Top N items & Categories

```sql
# 4 B. Category-level summary
SELECT
  Category,
  COUNT(*) AS tx_count,
  SUM(Purchase_Amount_USD) AS total_revenue,
  AVG(Purchase_Amount_USD) AS avg_amount,
  SUM(CASE WHEN Discount_Applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*) AS discount_rate,
  SUM(CASE WHEN Promo_Code_Used ='Yes' THEN 1 ELSE 0 END)/COUNT(*) AS promo_rate
FROM shopping_table
GROUP BY Category
ORDER BY total_revenue DESC;
```

Category	tx_count	total_revenue	avg_amount	discount_rate	promo_rate
Clothing	1737	104264	60.0253	0.4208	0.4208
Accessories	1240	74200	59.8387	0.4379	0.4379
Footwear	599	36093	60.2554	0.4324	0.4324
Outerwear	324	18524	57.1728	0.4444	0.4444

Result 28

Output

Action Output

#	Time	Action	Message
31	20:33:07	SELECT SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS age_null, SUM(CASE WHEN Gender...	1 row(s) returned
32	20:35:43	SELECT Category, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS total_revenue, AVG(Purch...	4 row(s) returned

```sql
# 5 C. Top N items and categories
SELECT Item_Purchased, Category,
       COUNT(*) AS tx_count,
       SUM(Purchase_Amount_USD) AS revenue
FROM shopping_table
GROUP BY Item_Purchased, Category
ORDER BY revenue DESC
LIMIT 20;

# 6 D. Discount effectiveness (compare avg amount with/without discount)
SELECT
```

Item_Purchased	Category	tx_count	revenue
Blouse	Clothing	171	10410
Shirt	Clothing	169	10332
Dress	Clothing	166	10320
Pants	Clothing	171	10090
Jewelry	Accessories	171	10010
Sunglasses	Accessories	161	9649
Belt	Accessories	161	9635
Scarf	Accessories	157	9561
Sweater	Clothing	164	9462
Shorts	Clothing	157	9433

Result 29

Output

Action Output

#	Time	Action
32	20:35:43	SELECT Category, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS total_revenue, AVG(Purch...
33	20:38:18	SELECT Item_Purchased, Category, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS reve...

5. Discount effectiveness:

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Limit to 50000 rows

```sql
# 6 D. Discount effectiveness (compare avg amount with/without discount)
SELECT
  Discount_Applied,
  COUNT(*) AS tx_count,
  AVG(Purchase_Amount_USD) AS avg_amount,
  SUM(Purchase_Amount_USD) AS total_revenue
FROM shopping_table
GROUP BY Discount_Applied;

# 7 E. Card spending by age group (create age-bins on-the-fly)
SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Discount_Applied	tx_count	avg_amount	total_revenue
No	2223	60.1305	133670
Yes	1677	59.2791	99411

Result 30

Output

Action Output

#	Time	Action
33	20:38:18	SELECT Item_Purchased, Category, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD
34	20:39:30	SELECT Discount_Applied, COUNT(*) AS tx_count, AVG(Purchase_Amount_USD) AS avg_amo

6. Card Spending by age group:

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Limit to 50000 rows

```sql
# 7 E. Card spending by age group (create age-bins on-the-fly)
SELECT
  CASE
    WHEN Age < 18 THEN '<18'
    WHEN Age BETWEEN 18 AND 24 THEN '18-24'
    WHEN Age BETWEEN 25 AND 34 THEN '25-34'
    WHEN Age BETWEEN 35 AND 49 THEN '35-49'
    WHEN Age BETWEEN 50 AND 64 THEN '50-64'
    ELSE '65+' END AS age_group,
  COUNT(*) AS tx_count,
  SUM(Purchase_Amount_USD) AS total_spent,
  AVG(Purchase_Amount_USD) AS avg_spent
FROM shopping_table
WHERE Payment_Method LIKE '%Card%'
GROUP BY age_group
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_group	tx_count	total_spent	avg_spent
18-24	157	9674	61.6178
25-34	263	15335	58.3080
35-49	391	23723	60.6726
50-64	378	22539	59.6270
65+	140	8414	60.1000

Result 31

Output

Action Output

#	Time	Action
34	20:39:30	SELECT Discount_Applied, COUNT(*) AS tx_count, AVG(Purchase_Amount_USI
35	20:40:58	SELECT CASE WHEN Age < 18 THEN '<18' WHEN Age BETWEEN 18 AND

7. Season Vs Location pivot (top combinations)

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Limit to 50000 rows

```
ORDER BY age_group;

# 8 F. Season x Location pivot (top combinations)
SELECT Season, Location,
       COUNT(*) AS tx_count,
       SUM(Purchase_Amount_USD) AS total_spend
FROM shopping_table
GROUP BY Season, Location
ORDER BY total_spend DESC
LIMIT 50;

# 9 G. Per-customer aggregated summary (RFM style + promo/discount behaviour)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Season	Location	tx_count	total_spend
Spring	Nevada	28	1916
Winter	Vermont	31	1885
Spring	Alaska	27	1848
Summer	Pennsylvania	27	1753
Winter	Virginia	25	1752
Fall	Texas	26	1732
Fall	California	27	1659
Summer	North Dakota	27	1656
Winter	Montana	25	1631

Result 32

Output

Action Output

#	Time	Action
35	20:40:58	SELECT CASE WHEN Age < 18 THEN '<18' WHEN Age BETWEEN 18 AND 24 THEN '18-24'
36	20:42:09	SELECT Season, Location, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS total_spe

8. Per-customer aggregated summary (RFM style + promo/discount behavior)

shopping trends analysis client p...

Limit to 50000 rows

```
LIMIT 50;

# 9 G. Per-customer aggregated summary (RFM style + promo/discount behaviour)
SELECT Customer_ID,
       COUNT(*) AS tx_count,
       SUM(Purchase_Amount_USD) AS total_spent,
       AVG(Purchase_Amount_USD) AS avg_spent,
       SUM(CASE WHEN Discount_Applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*) AS d
       SUM(CASE WHEN Promo_Code_Used = 'Yes' THEN 1 ELSE 0 END)/COUNT(*) AS pr
       MAX(Season) AS last_known_season,
       MAX(Location) AS top_location
FROM shopping_table
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	tx_count	total_spent	avg_spent	discount_rate	promo_rate	last_known_season	top_lo
1	1	53	53.0000	1.0000	1.0000	Winter	Kentuck
2	1	64	64.0000	1.0000	1.0000	Winter	Maine
3	1	73	73.0000	1.0000	1.0000	Spring	Massad
4	1	90	90.0000	1.0000	1.0000	Spring	Rhode I
5	1	49	49.0000	1.0000	1.0000	Spring	Oregon
6	1	20	20.0000	1.0000	1.0000	Summer	Wyomir
7	1	85	85.0000	1.0000	1.0000	Fall	Montan
8	1	34	34.0000	1.0000	1.0000	Winter	Louisian
9	1	97	97.0000	1.0000	1.0000	Summer	West Vi

Result 33

Output

Action Output

#	Time	Action
36	20:42:09	SELECT Season, Location, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS total_spe
37	20:43:14	SELECT Customer_ID, COUNT(*) AS tx_count, SUM(Purchase_Amount_USD) AS total_spent,

9. Subscription impact (subscriber Vs non-subcribers)

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Limit to 50000 rows

```
GROUP BY Customer_ID;

# 10 H. Subscription impact (subscribers vs non-subscribers)
SELECT Subscription_Status,
       COUNT(*) AS tx_count,
       SUM(Purchase_Amount_USD) AS total_spend,
       AVG(Purchase_Amount_USD) AS avg_spend
FROM shopping_table
GROUP BY Subscription_Status;

# 11. I. Payment method distribution
SELECT Payment_Method,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Subscription_Status	tx_count	total_spend	avg_spend
No	2847	170436	59.8651
Yes	1053	62645	59.4919

Result 34

Output

Action Output

#	Time	Action
37	20:43:14	SELECT Customer_ID, COUNT(*) AS tx_count, SUM(Purchase_Amount
38	20:44:27	SELECT Subscription_Status, COUNT(*) AS tx_count, SUM(Purchase_A

10. Payment method distribution :

shopping trends analysis client p...

Limit to 50000 rows

```sql
# 11. I. Payment method distribution
SELECT Payment_Method,
       COUNT(*) AS tx_count,
       SUM(Purchase_Amount_USD) AS total_spend,
       AVG(Purchase_Amount_USD) AS avg_spend
FROM shopping_table
GROUP BY Payment_Method
ORDER BY tx_count DESC;

# 12 J. Export subset for Python (recommended colu
SELECT Customer_ID, Age, Gender, Item_Purchased, C
```

Result Grid | Filter Rows: | Export: | Wrap Cell

Payment_Method	tx_count	total_spend	avg_spend
Credit Card	696	42567	61.1595
Venmo	653	39991	61.2420
Cash	648	38833	59.9275
PayPal	638	37449	58.6975
Debit Card	633	37118	58.6382
Bank Transfer	632	37123	58.7389

Result 35

Output

Action Output

#	Time	Action
38	20:44:27	SELECT Subscription_Status, COUNT(*) AS tx_count, S
39	20:45:11	SELECT Payment_Method, COUNT(*) AS tx_count, SUI